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Designs Related Through Projective and Hopf Maps

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Abstract
We verify a construction which, for $\mathbb{K}$ the reals, complex numbers, quaternions, or octonions, builds a spherical t-design by placing a spherical t-design on each $\mathbb{K}$-projective or $\mathbb{K}$-Hopf fiber associated to the points of a $\lfloor t/2 \rfloor$-design on a quotient projective space $\mathbb{KP}^n \neq \mathbb{OP}^2$ or sphere. This generalizes work of König and Kuperberg, who verified the $\mathbb{K} = \mathbb{C}$ case of the projective settings, and of Okuda, who (inspired by independent observation of this construction by Cohn, Conway, Elkies, and Kumar) verified the $\mathbb{K} = \mathbb{C}$ case of the generalized Hopf settings.

Keywords Spherical t-design · Projective t-design · Cubature · Quadrature · Projective map · Hopf map

Mathematics Subject Classification 05B30 (primary); 41A55 · 41A63 · 52C35 · 65D32 (secondary)

1 Background and main results

Spherical t-designs are finite subsets of spheres introduced by Delsarte, Goethals, and Seidel [10, Section 5] which provide good global approximations of the spheres they lie on, in the sense that polynomials of degree at most t have the same average value on a t-design as on the sphere. t-designs on the d-dimensional unit sphere S^d in $\mathbb{R}^{d+1}$ were shown to exist for all t and d by Seymour and Zaslavsky [27, Corollary 1] and later explicitly constructed by Bajnok as the direct product of $d-2$ weighted interval t-designs (i.e. finite subsets of the weighted interval which exactly average polynomials of degree at most t) and a regular $(t+1)$-gon [3]. Regular $(t+1)$-gons are known to be the only optimally small t-designs on S^1 for all t [10, Example 5.1.4], but in any other dimension, only a handful of optimally small spherical t-designs are known.

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It was only in the past decade that a landmark paper of Bondarenko, Radchenko, and Viazovska resolved the longstanding open problem of finding the asymptotic order as $t \to \infty$ of optimally small t-designs on S^d for $d > 1$. Specifically, these authors proved that there exist t-designs on S^d comprised of N points for any t, d, and $N \geq C_d t^d$ (C_d a constant depending on d) [6, Theorem 1], showing that lower bounds [10, Theorems 5.11, 5.12] on the sizes of spherical t-designs provided by Delsarte, Goethals, and Seidel are asymptotically optimal as $t \to \infty$. Considerable exploration of the properties of spherical designs—much of which is presented in a survey article by Bannai and Bannai [4]—has been done past these results since their introduction.
Projective t-designs are finite subsets of projective spaces which, in analogy with spherical t-designs, exactly average degree at most t polynomials on the projective spaces they lie on. These objects were first considered when Neumaier [24] introduced designs on *Delsarte spaces*, a class of metric spaces including projective spaces. As in the case of spherical designs, it follows from the work of Seymour and Zaslavsky [27, Main Theorem] that t-designs exist on any projective space for any t. Following the results of Bondarenko, Radchenko, and Viazovska in the spherical setting [6, Theorem 1], Etayo, Marzo, and Ortega-Cerdà proved that there exist t-designs on any d-dimensional algebraic manifold M (so notably, on any projective space of real dimension d) comprised of N points for any t, d, and $N \geq C_M t^d$ (C_M a constant depending on M) [13, Theorem 2.2], showing that lower bounds of Dunkl [11] and of Bannai and Hoggar [5, Theorem 1] on the sizes of t-designs on complex and quaternionic projective spaces analogous to the bounds of Delsarte, Goethals, and Seidel [10, Theorems 5.11, 5.12] in the spherical setting are asymptotically optimal as $t \to \infty$.
Spherical and complex projective designs can be related by the construction introduced by König [19, Corollary 1] and further investigated by Kuperberg [20, Theorem 4.1] which builds a t-design on S^{2n+1} by placing the vertices of a regular $(t+1)$-gon on each fiber of the complex projective map associated to a $\lfloor t/2 \rfloor$-design on $\mathbb{CP}^n$. A related construction which builds a t-design on S^3 by placing the vertices of a regular $(t+1)$-gon on each fiber of the Hopf map associated to a $\lfloor t/2 \rfloor$-design on S^2 was—inspired by the independent observation of the construction of König by Cohn, Conway, Elkies, and Kumar [8, Section 4]—verified by Okuda [25, Theorem 1.1]. We generalize these results to prove Theorem 1.1, whose main consequence is that one can build a spherical t-design by placing a spherical t-design on each fiber of the projective or generalized Hopf map associated to a $\lfloor t/2 \rfloor$-design on a quotient real, complex, quaternionic, or octonionic projective space $\mathbb{KP}^n \neq \mathbb{OP}^2$ or sphere. This gives new constructions of t-designs on S^{4n+3} from $\lfloor t/2 \rfloor$-designs on $\mathbb{HP}^n$ and t-designs on S^3 for any $n \in \mathbb{N}_+$, t-designs on S^7 from $\lfloor t/2 \rfloor$-designs on S^4 and t-designs on S^3, and t-designs on S^{15} from $\lfloor t/2 \rfloor$-designs on $\mathbb{OP}^1$ or S^8 and t-designs on S^7.

Theorem 1.1 (Main Theorem) *For $\mathbb{K}$ the reals $\mathbb{R}$, complex numbers $\mathbb{C}$, quaternions $\mathbb{H}$, or octonions $\mathbb{O}$ and $k := \dim_{\mathbb{R}} \mathbb{K} - 1$, take $\Pi : S^d \to \Sigma$ to be either the $\mathbb{K}$-projective map*

$$\Pi_{\mathbb{K}} : S^{(k+1)n+k} \subset \mathbb{K}^{n+1} \to \mathbb{KP}^n, \quad \omega \mapsto [\omega] \tag{1}$$

for $n \in \mathbb{N}$ ($n = 1$ if $\mathbb{K} = \mathbb{O}$) or the $\mathbb{K}$-Hopf map

$$\pi_{\mathbb{K}} : S^{2k+1} \subset \mathbb{K}^2 \to S^{k+1} \subset \mathbb{R} \times \mathbb{K}, \quad (a, b) \mapsto (|a|^2 - |b|^2, 2a\overline{b}). \tag{2}$$

For weighted 0-designs (Y, λ_Y) on Σ and (Z_y, λ_y) on S^k alongside base points $z_y \in \Pi^{-1}(y)$ for each $y \in Y$, we consider

$$X := \bigcup_{y \in Y} \{z_y z \mid z \in Z_y\}, \quad \lambda(z_y z) := \lambda_Y(y)\lambda_y(z) \ \text{ for } \ z \in Z_y. \tag{3}$$

For any $t \in \mathbb{N}$, (Y, λ_Y) is a weighted $\lfloor t/2 \rfloor$-design on Σ if (X, λ) is a weighted t-design on S^d. Moreover, (X, λ) is a weighted t-design on S^d if (Y, λ_Y) is a weighted $\lfloor t/2 \rfloor$-design on Σ and (Z_y, λ_y) is a weighted t-design on S^k for each $y \in Y$. If $|Z_y|$ is constant in $y \in Y$, the corresponding statements for unweighted designs (weighted designs with constant weight function) hold.

An interesting consequence of two applications of Theorem 1.1 is that, for $\mathbb{L} \subset \mathbb{K}$ the reals, complex numbers, or quaternions and $l := \dim_{\mathbb{R}} \mathbb{L} - 1$, we can build a t-design on $\mathbb{LP}^{(d-l)/(l+1)}$ by placing a t-design on $\Pi_{\mathbb{L}}(w) \cong \mathbb{LP}^{(k-l)/(l+1)}$ for each point $w \subset S^d$ of a t-design on $\mathbb{KP}^n$ as in (3).

We define the objects of interest in Subsection 1.1. We then prove the projective settings of Theorem 1.1 in Section 2 and use these results to verify the generalized Hopf settings of the theorem in Subsection 2.1. Finally, we provide examples—some of which are known to be optimally small—of unweighted spherical t-designs built as in Theorem 1.1 and discuss the sense in which the constructions of the theorem are each asymptotically optimal as $t \to \infty$ in Section 3.

1.1 Spaces, maps, and designs

Consider the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$, the quaternions $\mathbb{H}$, and the octonions $\mathbb{O}$, which Hurwitz's theorem [12, Section 10] states are exactly the normed division algebras over the real numbers. These algebras are respectively generated over $\mathbb{R}$ by elements $\{e_0\}$, $\{e_0, e_1\}$, $\{e_0, e_1, e_2, e_3\}$, and $\{e_i\}_{i=0}^7$ satisfying $e_0^2 = e_0$, $e_i^2 = -e_0$ for $i \neq 0$, and the relations illustrated by the Fano plane in Fig. 1. While $\mathbb{R}$ and $\mathbb{C}$ are commutative and associative, $\mathbb{H}$ is associative but noncommutative, and $\mathbb{O}$ is noncommutative and nonassociative but *alternative*—i.e., for all $o_1, o_2 \in \mathbb{O}$, $o_1(o_1 o_2) = (o_1 o_1)o_2$ and $o_1(o_2 o_2) = (o_1 o_2)o_2$ [2].

Fix $m \in \mathbb{N}$. By $\mathbb{R}^{m+1}$, $\mathbb{C}^{m+1}$, $\mathbb{H}^{m+1}$, and $\mathbb{O}^{m+1}$, we respectively denote the $(m+1)$-dimensional real, complex, quaternionic, and octonionic vector spaces. S^m is taken to be the m-dimensional unit sphere in $\mathbb{R}^{m+1}$ and is equipped with the uniform measure (meaning a measure on a metric space with respect to which any two balls of the same radius have the same measure) σ^m, which we abbreviate by σ and normalize such that $\sigma(S^m) = 1$. Then, picking

$$\begin{gathered} \mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}, \quad k := \dim_{\mathbb{R}}(\mathbb{K}) - 1, \\ n \in \mathbb{N} \quad (n = 1 \ \text{ if } \ \mathbb{K} = \mathbb{O}), \quad d := (k+1)n + k, \end{gathered} \tag{4}$$

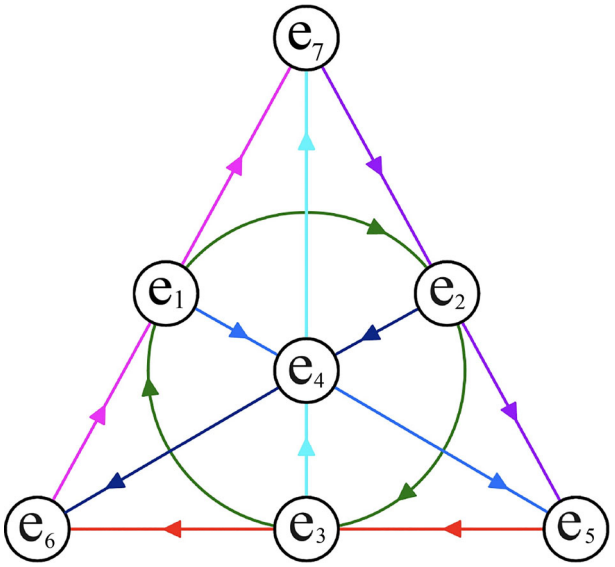


Fig. 1 The Fano plane, which visualizes how to multiply octonions. We consider 7 lines in this picture: the 3 sides of the triangle, its 3 altitudes, and the circle inscribed in the triangle. If arrows point from e_i to e_j to e_l along any one of these lines, we have $e_i e_j = -e_j e_i = e_l$

S^d is the unit sphere in $\mathbb{K}^{n+1}$. We define the $(k+1)n$-dimensional (over $\mathbb{R}$) $\mathbb{K}$-projective space $\mathbb{KP}^n$—the space of $\mathbb{K}$-lines in $\mathbb{K}^{n+1}$—using a right action of multiplication, setting

$$\mathbb{KP}^n := \{[\omega] \mid \omega \in S^d\}, \quad [\omega] := \begin{cases} \left\{ \frac{(\omega\overline{\omega}_{n+1})\zeta}{|\omega_{n+1}|} \;\middle|\; \zeta \in S^k \right\} & \omega_{n+1} \neq 0 \\ \{\omega\zeta \mid \zeta \in S^k\} & \omega_{n+1} = 0 \end{cases}. \tag{5}$$

Note that when $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, we may use the simpler presentation

$$[\omega] = \{\omega\zeta \mid \zeta \in S^k\}$$

by associativity of these algebras. We equip $\mathbb{KP}^n$ with the uniform measure $\rho^n_{\mathbb{K}}$, which we abbreviate by ρ and normalize such that $\rho(\mathbb{KP}^n) = 1$.
Our definition of $\mathbb{OP}^1$ is equivalent to the more common presentation

$$\left\{ \begin{bmatrix} a \\ b \end{bmatrix} \begin{bmatrix} \overline{a} & \overline{b} \end{bmatrix} \;\middle|\; (a, b) \in S^{15} \subset \mathbb{O}^2 \right\}$$

of $\mathbb{OP}^1$ discussed by Baez [2, Subsection 3.1]. Specifically, we can observe that $|a|^2 = |\alpha|^2$, $a\overline{b} = \alpha\overline{\beta}$, and $|b|^2 = |\beta|^2$ exactly when $[(a, b)] = [(\alpha, \beta)]$ for $(a, b), (\alpha, \beta) \in S^{15}$ using the fact that the subalgebra generated by any two elements of $\mathbb{O}$ is associative, as can be seen from alternativity of $\mathbb{O}$ combined with a lemma of Artin [26, Section 3]. Baez also discusses why it is sensible to consider octonionic projective spaces $\mathbb{OP}^n$ for

exactly $n \in \{1, 2\}$. The proofs in Section 2 do not directly apply to give a construction which builds a t-design on S^{23} from a $\lfloor t/2 \rfloor$-design on $\mathbb{OP}^2$ and t-designs on S^7 because $\mathbb{OP}^2$ does not admit a presentation analogous to (5). Instead, they directly verify a construction which uses a map $\Pi_{\mathbb{O}}$ analogous to (1) from the set $S(\mathbb{O}^3_A)$ of triples $(a, b, c) \in S^{23} \subset \mathbb{O}^3$ satisfying $(ab)c = a(bc)$ to $\mathbb{OP}^2$ to relate t-designs on $S(\mathbb{O}^3_A)$ to $\lfloor t/2 \rfloor$-designs on $\mathbb{OP}^2$ and t-designs on preimages of $\Pi_{\mathbb{O}}$.

Consider the $\mathbb{K}$-projective maps (1) and $\mathbb{K}$-Hopf map (2) as in Theorem 1.1. The measure ρ we consider on $\mathbb{KP}^n$ is equivalent to the pushforward $(\Pi_{\mathbb{K}})_*\sigma = \sigma \circ \Pi_{\mathbb{K}}^{-1}$ of the measure σ we consider on S^d by the $\mathbb{K}$-projective map, as can be seen since $(\Pi_{\mathbb{K}})_*\sigma$ will be uniform by symmetry and will be normalized such that $((\Pi_{\mathbb{K}})_*\sigma)(\mathbb{KP}^n) = \sigma(S^d) = 1$. Additionally, we have $\pi_{\mathbb{K}} = h_{\mathbb{K}} \circ \Pi_{\mathbb{K}}$, where

$$h_{\mathbb{K}} : \mathbb{KP}^1 \to S^{k+1} \subset \mathbb{R} \times \mathbb{K}, \quad [(a, b)] \mapsto (|a|^2 - |b|^2, 2a\overline{b}) \tag{6}$$

is a diffeomorphism identifying $\mathbb{KP}^1$ with S^{k+1}. We note that $\sigma = (h_{\mathbb{K}})_*\rho$, so we also have $\sigma^{k+1} = (\pi_{\mathbb{K}})_*\sigma^{2k+1}$. Preimages of the $\mathbb{K}$-projective and $\mathbb{K}$-Hopf maps can be seen from (1) to each arise as the unit sphere in a subspace of $\mathbb{K}^{n+1}$ of dimension 1 over $\mathbb{K}$. We equip these fibers with the uniform measure σ, which we normalize such that each fiber has volume 1.

The space of real-valued polynomials on $\mathbb{R}^{m+1}$ is the closure of the constant and coordinate functions $\mathbb{R}^{m+1} \to \mathbb{R}$ under addition and multiplication. We then say that $P_t(\mathbb{R}^{m+1})$ is the space of polynomials of degree less than or equal to t on $\mathbb{R}^{m+1}$; i.e. polynomials whose terms are each the product of a constant function and at most t coordinate functions. We then define

$$P_t(S^m) := \{f|_{S^m} \mid f \in P_t(\mathbb{R}^{m+1})\}, \tag{7}$$

$$P_t(\mathbb{KP}^n) := \{f : \mathbb{KP}^n \to \mathbb{R} \mid \Pi_{\mathbb{K}}^* f \in P_{2t}(S^d)\}, \tag{8}$$

where $\Pi_{\mathbb{K}}^* f$ is the pullback $f \circ \Pi_{\mathbb{K}}$ of f by $\Pi_{\mathbb{K}}$. We consider coordinates in $\mathbb{K}$ for real-valued polynomials via the association $\mathbb{K} \cong \mathbb{R}^{k+1}$ which realizes $\mathbb{K}$ as a $(k+1)$-dimensional real vector space with basis vectors $\{e_i\}_{i=0}^k$, the standard generators for the algebra $\mathbb{K}$ over $\mathbb{R}$.

We do not use explicit spanning sets of polynomials on projective spaces in the proofs below, but note that Lyubich and Shatalova [23, Main Theorem] proved that for $\mathbb{K} \in \{\mathbb{C}, \mathbb{H}\}$ and where $\langle a, b\rangle = \sum_{i=1}^{n+1} a_i \overline{b_i}$ is the conventional inner product on $\mathbb{K}^{n+1}$, $P_t(\mathbb{KP}^n)$ is spanned over $\mathbb{K}$ by the functions

$$\left\{[\omega_1] \mapsto |\langle \omega_1, \omega_2\rangle|^s \;\middle|\; [\omega_2] \in \mathbb{KP}^n,\ s \in \{0, 2, \ldots, 2t\}\right\}.$$

We now introduce weighted and unweighted t-designs, numerical integration formulas (also known as *quadrature* or *cubature rules*) which precisely average polynomials of degree t or less.

Definition 1.2 Take (Ω, μ) to be a sphere (S^m, σ) or projective space $(\mathbb{KP}^n, \rho)$, so we note $\mu(\Omega) = 1$. The pair (X, λ) of a finite subset $X \subset \Omega$ and a function $\lambda : X \to$

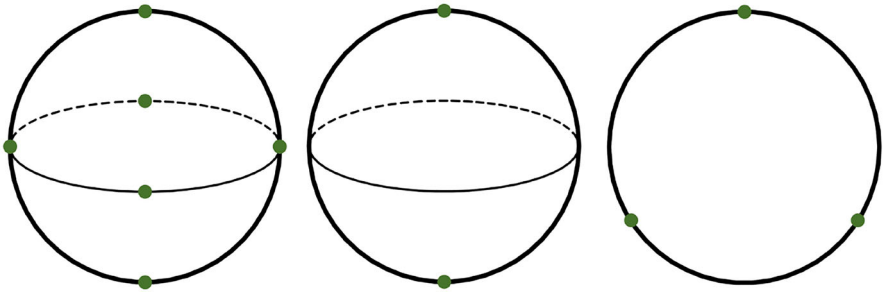


Fig. 2 Left: vertices of an octahedron, a 3-design on S^2. Center: antipodal points, a 1-design on S^2. Right: vertices of a triangle, a 2-design on S^1

$\mathbb{R}_+ := \{r \in \mathbb{R} \mid r > 0\}$ is called a *weighted t-design* on Ω if

$$\sum_{x \in X} \lambda(x) f(x) = \int_\Omega f \, d\mu \quad \text{for all} \quad f \in P_t(\Omega).$$

We may call (X, λ) a *weighted spherical t-design* if Ω is a sphere and a *weighted* $\mathbb{K}$*-projective t-design* if Ω is a $\mathbb{K}$-projective space. Additionally, X is called an *unweighted t-design* (or simply a *t-design*) if λ is the constant function $|X|^{-1}$. Examples of such unweighted t-designs are presented in Fig. 2.

Literature discussing projective designs often defines these objects as finite subsets of projective spaces on which the averages of certain Jacobi polynomials vanish. It was proven by Lyubich [22, Proposition 2.2] that the projective settings of Definition 1.2 are equivalent to the definition more common in the literature. We note that the work of Lyubich has a typo: the word "tight" in their proposition should be omitted.

2 Validity of the constructions

In this section, we prove Theorem 1.1. We first verify Theorem 1.1 when Π is a projective map, then show in Subsection 2.1 that the $n = 1$ cases of the projective settings give rise to the results of Theorem 1.1 when Π is a generalized Hopf map.

Proof of the projective settings of Theorem 1.1 Taking $\mathbb{K}, k, n$ and d as in (4) along with $t \in \mathbb{N}$, we consider weighted 0-designs (Y, λ_Y) on $\mathbb{KP}^n$ and (Z_y, λ_y) on S^k alongside base points $z_y \in y$ for each $y \in Y$. We then define (X, λ) as in (3).

Pick $g \in P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n)$. Noting that $(\Pi_{\mathbb{K}}^* g)(z_y z) = g(y)$ for any $z \in Z_y$ and that (Z_y, λ_y) is a 0-design on S^k for each $y \in Y$, we have

$$\begin{aligned} \sum_{y \in Y} \sum_{z \in Z_y} \lambda_Y(y) \lambda_y(z) (\Pi_{\mathbb{K}}^* g)(z_y z) &= \sum_{y \in Y} \sum_{z \in Z_y} \lambda_Y(y) \lambda_y(z) g(y) \\ &= \sum_{y \in Y} \lambda_Y(y) g(y). \end{aligned} \tag{9}$$

Additionally, we may observe from the definition (22) of $I_{\mathbb{K}}$ that

$$g = (I_{\mathbb{K}} \circ \Pi^*_{\mathbb{K}})g. \tag{10}$$

Combining (9) and (10) with Definition 1.2, we see that (Y, λ_Y) is a weighted $\lfloor t/2 \rfloor$-design on $\mathbb{KP}^n$ exactly when

$$\sum_{y\in Y}\sum_{z\in Z_y} \lambda_Y(y)\lambda_y(z) f(z_y z) = \int_{\mathbb{KP}^n} I_{\mathbb{K}} f \, d\rho \tag{11}$$

for all $f \in \Pi^*_{\mathbb{K}}\big(P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n)\big)$. Now, we also see from Definition 1.2 combined with the result

$$\int_{\mathbb{KP}^n} I_{\mathbb{K}} f \, d\rho = \int_{S^d} f \, d\sigma \quad \textit{for} \quad f \in L^1(S^d)$$

of Lemma 2.2 that (11) being satisfied for $f \in P_t(S^d)$ is exactly the condition that (X, λ) is a weighted t-design on S^d. Therefore, since we can see from the definition (8) of polynomials on projective spaces that

$$\Pi^*_{\mathbb{K}}\Big(P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n)\Big) \subseteq P_t(S^d),$$

we have shown that (Y, λ_Y) is a weighted $\lfloor t/2 \rfloor$-design on $\mathbb{KP}^n$ if (X, λ) is a weighted t-design on S^d.

Now, say (Z_y, λ_y) is a weighted t-design on S^k for each $y \in Y$. Picking $f \in P_t(S^d)$, (27) in Lemma 2.3 shows that $f(z_y\zeta) \in P_t(S^k)$ for $y \in Y$. So, applying a change of variables $z_y\zeta \mapsto \zeta$ using (19) followed by the definition (22) of $I_{\mathbb{K}}$, we have that

$$\sum_{z\in Z_y} \lambda_y(z) f(z_y z) = \int_{S^k} f(z_y\zeta)\, d\sigma(\zeta) = \int_y f(\zeta)\, d\sigma(\zeta) = (I_{\mathbb{K}} f)(y)$$

for $y \in Y$, which shows that

$$\sum_{y\in Y}\sum_{z\in Z_y} \lambda_Y(y)\lambda_y(z) f(z_y z) = \sum_{y\in Y} \lambda_Y(y)(I_{\mathbb{K}} f)(y). \tag{12}$$

From (28) in Lemma 2.3, we see that $I_{\mathbb{K}} f \in P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n)$, so combining (12) with (9) for $g = I_{\mathbb{K}} f$ gives us that

$$\sum_{y\in Y}\sum_{z\in Z_y} \lambda_Y(y)\lambda_y(z) f(z_y z) = \sum_{y\in Y}\sum_{z\in Z_y} \lambda_Y(y)\lambda_y(z)((\Pi^*_{\mathbb{K}} \circ I_{\mathbb{K}}) f)(z_y z). \tag{13}$$

Additionally, we observe from (10) that

$$\int_{\mathbb{KP}^n} (I_{\mathbb{K}} \circ \Pi^*_{\mathbb{K}} \circ I_{\mathbb{K}}) f \, d\rho = \int_{\mathbb{KP}^n} I_{\mathbb{K}} f \, d\rho. \tag{14}$$

From (13) and (14), we see that (11) is satisfied for $f \in P_t(S^d)$ if it is satisfied for

$$(\Pi_{\mathbb{K}}^* \circ I_{\mathbb{K}}) f \in \Pi_{\mathbb{K}}^*\Big(P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n)\Big),$$

so (X, λ) is a weighted t-design on S^d if (Y, λ_Y) is a weighted $\lfloor t/2 \rfloor$-design on $\mathbb{KP}^n$. □

Elements of the proof of the projective settings of Theorem 1.1 were inspired by work of Okuda [25, Lemma 3.9] verifying the $\mathbb{C}$-Hopf setting of Theorem 1.1. We now prove facts cited in this proof. With $\mathbb{K}$, k, n, and d as in (4), consider the left multiplication by a base point $z_w \in w$ map

$$\zeta_w : S^k \to w, \quad \zeta \mapsto z_w \zeta \tag{15}$$

we define for each $w \in \mathbb{KP}^n$. Taking

$$U := \{[\omega] \in \mathbb{KP}^n \mid \omega_{n+1} \neq 0\} \tag{16}$$

along with

$$V := \Pi_{\mathbb{K}}^{-1}(U) = \{\omega \in S^d \mid \omega_{n+1} \neq 0\} \tag{17}$$

and respectively equipping these spaces with the restrictions of the measures ρ and σ discussed in Subsection 1.1, we consider a local trivialization

$$\zeta_U : U \times S^k \to V, \quad ([\omega], \zeta) \mapsto \frac{(\omega\overline{\omega}_{n+1})\zeta}{|\omega_{n+1}|} \tag{18}$$

of the $\mathbb{K}$-projective map, where we equip $U \times S^k$ with the product measure $\mu := \rho \times \sigma$. Observe that ζ_w pushes forward the uniform measure on S^k to the uniform measure on $w \cong S^k$; i.e.

$$(\zeta_w)_* \sigma = \sigma \quad \textit{for any} \quad w \in \mathbb{KP}^n. \tag{19}$$

We now discuss how ζ_U pushes forward the product measure μ on $U \times S^k$ to the restriction of the uniform measure σ on S^d to V:

Lemma 2.1 *We have*

$$(\zeta_U)_* \mu = \sigma. \tag{20}$$

Proof (20) will be satisfied exactly when

$$(\zeta_U^{-1})_*(\sigma) = \mu = \rho \times \sigma. \tag{21}$$

We can see that

$$\zeta_U^{-1} : V \to U \times S^k, \quad \omega \mapsto \left([\omega], \frac{\omega_{n+1}}{|\omega_{n+1}|}\right) =: (\zeta_1^{-1}, \zeta_2^{-1})(\omega),$$

so we have

$$(\zeta_1^{-1})_*\sigma = \rho, \quad (\zeta_2^{-1})_*(\sigma^d) = \sigma^k,$$

as both pushforward measures are uniform by symmetry and normalized such that

$$((\zeta_1^{-1})_*\sigma)(U) = 1 = \rho(U), \quad ((\zeta_2^{-1})_*\sigma)(S^k) = 1 = \sigma(S^k).$$

This exactly shows that (21) is satisfied. □

Similarly to as was done by Okuda [25, Lemma 4.2], we use (19) and (20) alongside Fubini's theorem [14] to show that, with $\mathbb{K}$, n, and d as in (4), taking the average of any Lebesgue integrable function f over S^d provides the same result as taking the average on $\mathbb{KP}^n$ of the function $(I_{\mathbb{K}} f)(w)$ which averages f over each $\mathbb{K}$-projective fiber w. For such f and w, recalling as in Subsection 1.1 that we normalize the measure σ on w such that $\sigma(w) = 1$, we have

$$(I_{\mathbb{K}} f)(w) = \int_w f \, d\sigma. \tag{22}$$

Lemma 2.2 *For $f \in L^1(S^d)$, we have*

$$\int_{\mathbb{KP}^n} I_{\mathbb{K}} f \, d\rho = \int_{S^d} f \, d\sigma. \tag{23}$$

Proof Pick $f \in L^1(S^d)$ and recall the definitions (16) and (17) of U and V respectively. We see using that $S^d \setminus V$ has measure zero in S^d followed by a change of variables using (20) in Lemma 2.1 that

$$\int_{S^d} f \, d\sigma = \int_V f \, d\sigma = \int_{U \times S^k} \zeta_U^* f \, d\mu. \tag{24}$$

Since $f \in L^1(S^d)$ and $\mu = \rho \times \sigma$ is the product measure on $U \times S^k$, Fubini's theorem [14] followed by the fact that, taking $z_w = \zeta_U(w, 1) \in w$ for each $w \in \mathbb{KP}^n$, $\zeta_w = \zeta_U(w, \cdot)$ shows that

$$\int_{U \times S^k} \zeta_U^* f \, d\mu = \int_U \int_{S^k} \zeta_U^* f \, d\sigma \, d\rho = \int_U \int_{S^k} \zeta_w^* f \, d\sigma \, d\rho(w). \tag{25}$$

Changing variables using (19) and applying the definition (22) of $I_{\mathbb{K}}$ alongside the fact that $\mathbb{KP}^n \setminus U$ has measure zero in $\mathbb{KP}^n$, we have

$$\int_U \int_{S^k} \zeta_w^* f \, d\sigma \, d\rho(w) = \int_U \int_w f \, d\sigma \, d\rho = \int_{\mathbb{KP}^n} (I_{\mathbb{K}} f) \, d\rho. \tag{26}$$

Combining (24), (25), and (26) then provides us with the desired result (23). □

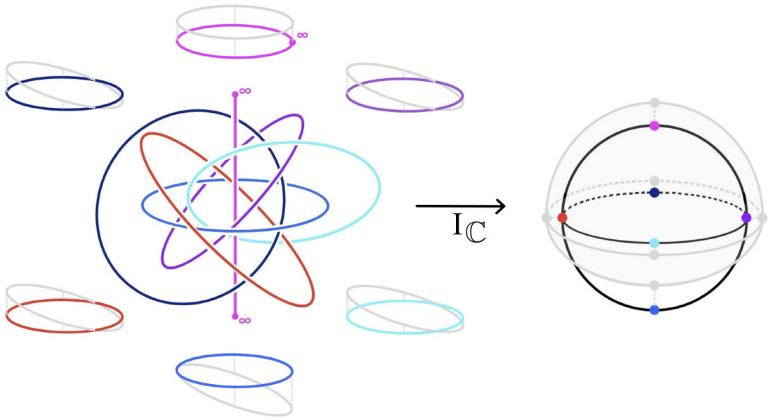


Fig. 3 Taking coordinates $(a, b = b_1 + ib_2) \in \mathbb{C}^2$ on $S^3 \cong \mathbb{R}^3 \cup \{\infty\}$ and $(\xi, \eta) \in \mathbb{R} \times \mathbb{C}$ on $\mathbb{CP}^1 \cong S^2$, restrictions of $f = |a|^2 - |b|^2 + 2|a|^2|b|^2 - b_1$ to projective fibers are pictured in gray on the left and $(I_{\mathbb{C}} f)(\xi, \eta) = \xi + \frac{1}{2}|\eta|^2$ is visualized in gray on the right, with points on the right denoted by the same color as their corresponding projective fibers on the left

Again consider $\mathbb{K}$, k, n, and d as in (4) along with $t \in \mathbb{N}$. We now discuss how $I_{\mathbb{K}}$ as in (22) and the pullback operator ζ_w^* induced by ζ_w as in (15) relate polynomials (as is visualized in Fig. 3).

Lemma 2.3 *We have*

$$\zeta_w^*\big(P_t(S^d)\big) = P_t(S^k) \quad \textit{for any} \quad w \in \mathbb{KP}^n, \tag{27}$$

$$I_{\mathbb{K}}\big(P_t(S^d)\big) = P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n). \tag{28}$$

Proof Fix $w \in \mathbb{KP}^n$ and consider

$$W := \{\lambda\omega \mid \lambda \in \mathbb{R},\ \omega \in w\} \cong \mathbb{R}^{k+1}.$$

Extending $\zeta_w : S^k \to w$ to the map $\zeta_w : \mathbb{R}^{k+1} \cong \mathbb{K} \to W$ taking $a \mapsto z_w a$, the definition of polynomials on $\mathbb{R}^{m+1}$ in Subsection 1.1 makes it clear that

$$\zeta_w^*\big(P_t(\mathbb{R}^{d+1})\big) = P_t(\mathbb{R}^{k+1}).$$

The definition (7) of polynomials on spheres then shows (27).
Note that the definition (22) of $I_{\mathbb{K}}$ shows that $I_{\mathbb{K}} \circ \Pi_{\mathbb{K}}^*$ is the identity. Combining the definition of $P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n)$ with this fact then shows that

$$P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n) = I_{\mathbb{K}}(\Pi_{\mathbb{K}}^*(P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n))) \subseteq I_{\mathbb{K}}(P_t(S^d)).$$

To prove (28), we must then only show that $I_{\mathbb{K}}(P_t(S^d)) \subseteq P_{\lfloor t/2 \rfloor}(\mathbb{KP}^n)$, and the definition (8) of polynomials on $\mathbb{KP}^n$ shows that this will be true if

$$\Pi_{\mathbb{K}}^*(I_{\mathbb{K}} f) \in P_{2\lfloor t/2 \rfloor}(S^d) \tag{29}$$

for all $f \in P_t(S^d)$. Pick such f to be a monomial of degree s. We can observe from the description of multiplication of elements of $\mathbb{K}$ in Subsection 1.1 that for any $\omega \in S^d \subset \mathbb{K}^{n+1}$ and $\zeta \in S^k \subset \mathbb{K}$,

$$f(\omega\zeta) = \sum_{i=1}^{(k+1)^s} g_i(\omega)h_i(\zeta),$$

where g_i and h_i are monomials of degree s on S^d and S^k respectively for all $i \in \{1, \ldots, (k+1)^s\}$. Therefore, applying a change of variables $\zeta \mapsto \omega\zeta$ for any base point $\omega \in w \in \mathbb{KP}^n$ using (19), we see that

$$(I_{\mathbb{K}} f)(w) = \int_w f(\zeta)\, d\sigma(\zeta) = \int_{S^k} f(\omega\zeta)\, d\sigma(\zeta) = \sum_{i=1}^{(k+1)^s} g_i(\omega) \int_{S^k} h_i(\zeta)\, d\sigma(\zeta).$$

Applying this fact, we see

$$(\Pi^*_{\mathbb{K}}(I_{\mathbb{K}} f))(\omega) = (I_{\mathbb{K}} f)[\omega] = \sum_{i=1}^{(k+1)^s} g_i(\omega) \int_{S^k} h_i(\zeta)\, d\sigma(\zeta) \in P_s(S^d).$$

Now, Lemma 2.4 shows that $\Pi^*_{\mathbb{K}}(I_{\mathbb{K}} f) = 0$ if $s = 2\lfloor t/2 \rfloor + 1$, so (29) is satisfied, completing the proof. □

Lemma 2.4 *For any polynomial $f \in P_{2t+1}(S^d)$ with terms having only odd degree, $I_{\mathbb{K}} f = 0$.*

Proof Say $f \in P_{2t+1}(S^d)$ is as in the statement of the lemma. We assume without loss of generality that f is a monomial of odd degree s. If $\mathbb{K} = \mathbb{R}$, we have

$$(I_{\mathbb{K}} f)([\omega]) = \frac{1}{2}(f(\omega) + f(-\omega)) = 0$$

for any $\omega \in S^d$. Now, take $\mathbb{K} \in \{\mathbb{C}, \mathbb{H}, \mathbb{O}\}$, so $k+1 = \dim_{\mathbb{R}} \mathbb{K}$ is even and thus $d+1 = (k+1)(n+1)$ is even for any $n \in \mathbb{N}$. The antipodal map on S^d has differential $-I_{d+1}$ (the negative of the identity on $\mathbb{R}^{d+1}$), so since

$$\det(-I_{d+1}) = (-1)^{d+1} = 1,$$

we see applying the change of variables $\zeta \mapsto -\zeta$ that

$$(I_{\mathbb{K}} f)(w) = \int_w f(\zeta)\, d\sigma(\zeta) = (-1)^s \int_w f(\zeta)\, d\sigma(\zeta) = -(I_{\mathbb{K}} f)(w)$$

because f is a monomial of odd degree s. So, we must have $I_{\mathbb{K}} f = 0$. □

2.1 The generalized Hopf settings

We show in this subsection that the results of Theorem 1.1 which use generalized Hopf maps to relate designs are equivalent to the $n = 1$ cases of the corresponding projective settings verified in Section 2. To this end, pick $\mathbb{K}$, k, and d as in (4) and recall the diffeomorphism $h_{\mathbb{K}}$ described in (6) which identifies $\mathbb{KP}^1$ with S^{k+1}.

Lemma 2.5 *We have*

$$h_{\mathbb{K}}^*\big(P_t(S^{k+1})\big) = P_t(\mathbb{KP}^1).$$

As Lemma 2.5 also implies that

$$P_t(S^{k+1}) = (h_{\mathbb{K}}^{-1})^*(P_t(\mathbb{KP}^1)),$$

the lemma combined with the fact that $(h_{\mathbb{K}})_*\mu = \sigma$ shows Corollary 2.6.

Corollary 2.6 *(Y, λ_Y) is a weighted t-design on $\mathbb{KP}^1$ if and only if $(h_{\mathbb{K}}(Y), (h_{\mathbb{K}}^{-1})^*\lambda_Y)$ is a weighted t-design on S^{k+1}.*

Applying Corollary 2.6, the $n = 1$ case of the $\mathbb{K}$-projective setting of Theorem 1.1 proves the corresponding $\mathbb{K}$-Hopf setting of the theorem.

Proof of Lemma 2.5 For any $(a, b) \in S^{2k+1}$, we note that

$$w := [(a, b)] = \pi_{\mathbb{K}}^{-1}(\pi_{\mathbb{K}}(a, b)).$$

Therefore, for any $g \in P_t(S^{k+1})$, we have from the definition (22) of $I_{\mathbb{K}}$ that

$$((I_{\mathbb{K}} \circ \pi_{\mathbb{K}}^*)g)(w) = \int_{\pi_{\mathbb{K}}^{-1}(\pi_{\mathbb{K}}(a,b))} \pi_{\mathbb{K}}^* g \, d\sigma = (\pi_{\mathbb{K}}^* g)(a, b).$$

We can therefore see from definitions (2) of $\pi_{\mathbb{K}}$ and (6) of $h_{\mathbb{K}}$ that

$$I_{\mathbb{K}} \circ \pi_{\mathbb{K}}^* = h_{\mathbb{K}}^*. \tag{30}$$

We also can observe from the description of multiplication of elements of $\mathbb{K}$ in Subsection 1.1 that

$$\pi_{\mathbb{K}}^*(P_t(S^{k+1})) \subseteq P_{2t}(S^{2k+1}). \tag{31}$$

Combining (30) with (31) and (28) in Lemma 2.3, we see that

$$h_{\mathbb{K}}^*(P_t(S^{k+1})) \subseteq I_{\mathbb{K}}(P_{2t}(S^{2k+1})) = P_t(\mathbb{KP}^1).$$

So, to complete the proof, we must only show that

$$h_{\mathbb{K}}^*\big(P_t(S^{k+1})\big) \supseteq P_t(\mathbb{KP}^1), \tag{32}$$

which will in turn be true if

$$\pi_{\mathbb{K}}^*(P_t(S^{k+1})) \supseteq \Pi_{\mathbb{K}}^*(P_t(\mathbb{KP}^1)) \tag{33}$$

since applying $I_{\mathbb{K}}$ to each side of (33) gives (32) by (30) and the fact we may observe from the definition (22) of $I_{\mathbb{K}}$ that $I_{\mathbb{K}} \circ \Pi_{\mathbb{K}}^*$ is the identity map. Fix $f \in \Pi_{\mathbb{K}}^*(P_t(\mathbb{KP}^1))$ and note that $f \in P_{2t}(S^{2k+1})$ by the definition (8) of $P_t(\mathbb{KP}^1)$, so we may consider $f_{\alpha,\beta} \in P_{2t}(S^{2k+1})$ and constants $c_{\alpha,\beta}$ for $\alpha, \beta \in \mathbb{N}$ satisfying $\alpha + \beta \leq 2t$ such that

$$f = \sum_{\alpha+\beta\leq 2t} f_{\alpha,\beta}, \quad |f_{\alpha,\beta}(a,b)| = c_{\alpha,\beta}|a|^\alpha |b|^\beta.$$

Again since $I_{\mathbb{K}} \circ \Pi_{\mathbb{K}}^*$ is the identity map, we see that $\Pi_{\mathbb{K}}^* \circ I_{\mathbb{K}}$ is the identity on $\Pi_{\mathbb{K}}^*(P_t(\mathbb{KP}^1))$. Also note that this operator is linear. Defining

$$\widetilde{f}_{\alpha,\beta} := (\Pi_{\mathbb{K}}^* \circ I_{\mathbb{K}}) f_{\alpha,\beta},$$

we therefore have that

$$f = (\Pi_{\mathbb{K}}^* \circ I_{\mathbb{K}}) f = \sum_{\alpha+\beta\leq 2t} \widetilde{f}_{\alpha,\beta}. \tag{34}$$

Now, for any α and β as above, we see from Lemma 2.7 that there exists $\widetilde{c}_{\alpha,\beta} \in [0, c]$ satisfying

$$|\widetilde{f}_{\alpha,\beta}(a,b)| = \widetilde{c}_{\alpha,\beta}|a|^\alpha |b|^\beta.$$

Assume without loss of generality that $\alpha \geq \beta$. We may observe from the definition (1) of $\Pi_{\mathbb{K}}$ that

$$\widetilde{f}_{\alpha,\beta}(a,b) = \widetilde{f}_{\alpha,\beta}\left(|a|, \frac{b\overline{a}}{|a|}\right) = |a|^{\alpha-\beta}\, \widetilde{f}_{\alpha,\beta}(1,0,\ldots,0,b\overline{a}).$$

Lemma 2.4 then guarantees that $\widetilde{f}_{\alpha,\beta}$ must have terms of only even degree, so we must have $\alpha - \beta = 2\gamma_{\alpha,\beta}$ for some $\gamma_{\alpha,\beta} \in \mathbb{N}$. Therefore,

$$\begin{aligned}\widetilde{f}_{\alpha,\beta}(a,b) &= |a|^{2\gamma_{\alpha,\beta}}\, \widetilde{f}_{\alpha,\beta}(1,0,\ldots,0,b\overline{a}) \\ &= \left(\frac{1}{2}(|a|^2 - |b|^2) + \frac{1}{2}\right)^{\gamma_{\alpha,\beta}} \widetilde{f}_{\alpha,\beta}(1,0,\ldots,0,\overline{a\overline{b}}).\end{aligned}$$

We thus have $\widetilde{f}_{\alpha,\beta} \in \pi_{\mathbb{K}}^*(P_{\alpha+\beta}(S^{k+1}))$ by the definition (2) of $\pi_{\mathbb{K}}$, showing by linearity of $\pi_{\mathbb{K}}^*$ that $f \in \pi_{\mathbb{K}}^*(P_t(S^{k+1}))$. □

Lemma 2.7 *For $\alpha, \beta \in \mathbb{N}$, $f \in P_{\alpha+\beta}(S^{2k+1})$, and $\widetilde{f} := (\Pi_{\mathbb{K}}^* \circ I_{\mathbb{K}}) f$, if*

$$|f(a,b)| = c|a|^\alpha |b|^\beta \tag{35}$$

for some $c \in \mathbb{R}$, we have

$$|\widetilde{f}(a,b)| = \widetilde{c}|a|^{\alpha}|b|^{\beta} \tag{36}$$

for some $\widetilde{c} \in [0, c]$.

Proof Consider f as in the statement of the lemma and label $w := [(a, b)]$. For any

$$z_w = (z_{w,1}, z_{w,2}) \in w,$$

we may apply a change of variables $\zeta \mapsto z_w\zeta$ using (19) to see that

$$|\widetilde{f}(a,b)| = \left|\int_w f(\zeta)\, d\sigma(\zeta)\right| = \left|\int_{S^k} f(z_w\zeta)\, d\sigma(\zeta)\right| \leq \int_{S^k} |f(z_w\zeta)|\, d\sigma(\zeta). \tag{37}$$

We then see from (35) that

$$\begin{aligned}\int_{S^k} |f(z_w\zeta)|\, d\sigma(\zeta) &= \int_{S^k} c|z_{w,1}\zeta|^{\alpha}|z_{w,2}\zeta|^{\beta}\, d\sigma(\zeta)\\ &= c|z_{w,1}|^{\alpha}|z_{w,2}|^{\beta}\int_{S^k} |\zeta|^{\alpha+\beta}\, d\sigma(\zeta)\\ &= c|z_{w,1}|^{\alpha}|z_{w,2}|^{\beta}.\end{aligned} \tag{38}$$

The definition (1) of $\Pi_{\mathbb{K}}$ and the fact that $z_w \in w = [(a, b)]$ then demonstrate that $|z_{w,1}| = |a|$ and $|z_{w,2}| = |b|$. Combining this with (37) and (38), we can see that

$$|\widetilde{f}(a,b)| \leq c|a|^{\alpha}|b|^{\beta}. \tag{39}$$

Every (nonzero) term of $\widetilde{f}(a, b)$ must thus contain at least α multiples of a or $\overline{a}$ and at least β multiples of b or $\overline{b}$, as (39) would otherwise not be satisfied when $a \to 0$ or $b \to 0$. Now, we can observe from the definition (8) of $P_t(\mathbb{KP}^1)$ and (28) in Lemma 2.3 that

$$\widetilde{f} = (\Pi_{\mathbb{K}}^* \circ I_{\mathbb{K}})f \in P_{\alpha+\beta}(S^{2k+1}),$$

so the degree of $\widetilde{f}$ does not exceed $\alpha + \beta$. Therefore, every (nonzero) term of $\widetilde{f}(a, b)$ must contain exactly α multiples of a or $\overline{a}$ and exactly β multiples of b or $\overline{b}$. This fact combined with (39) completes the proof. □

3 Examples and optimality

We now present examples of t-designs on S^d which can be built using the constructions communicated by Theorem 1.1, remark on the sizes of such t-designs, and discuss the sense in which these constructions are asymptotically optimal as $t \to \infty$. Note that all examples provided using the complex Hopf or complex projective settings could also have been produced using the results of Okuda [25, Theorem 1.1] or König [19, Corollary 1] and Kuperberg [20, Theorem 4.1] respectively.

We use the fact that, for $N \geq t+1$, the vertices

$$V_N := \left\{ e^{2m\pi i/N} \mid m \in \{0, \ldots, N-1\} \right\} \tag{40}$$

of a regular N-gon are a t-design on S^1 [10, Example 5.1.4]. We also note that we pick base points for the fibers of the $\mathbb{K}$-Hopf maps using the equation

$$\pi_{\mathbb{K}}^{-1}(\xi, \eta) = \begin{cases} \left\{ \frac{1}{\sqrt{2}} \left(\zeta \sqrt{1+\xi}, \frac{\overline{\eta}\zeta}{\sqrt{1+\xi}} \right) \,\middle|\, \zeta \in S^k \right\}, & \xi \neq -1 \\ \{(0, \zeta) \mid \zeta \in S^k\}, & \xi = -1 \end{cases}. \tag{41}$$

We first discuss an example remarked on by Okuda [25, Example 2.4].

Example 3.1 The set $A = \{(\pm 1, 0)\}$ of poles in $S^2 \subset \mathbb{R} \times \mathbb{C}$ is a 1-design. We also know that the vertices V_3 of an equilateral triangle are a 2-design on S^1. We see from (41) that

$$\pi_{\mathbb{C}}^{-1}(1,0) = \{(\zeta, 0) \mid \zeta \in S^1\}, \quad \pi_{\mathbb{C}}^{-1}(-1,0) = \{(0, \zeta) \mid \zeta \in S^1\},$$

so the $\mathbb{C}$-Hopf setting of Theorem 1.1 gives us that the set

$$X := \left\{ \left(z_1 e^{2m\pi i/3}, 0\right), \left(0, z_2 e^{2m\pi i/3}\right) \,\middle|\, m \in \{0,1,2\} \right\} \subset S^3 \subset \mathbb{C}^2$$

is a 2-design on S^3 for any $z_1, z_2 \in S^1$. Specifically, we see that

$$\left\{ (1,0), \left(e^{2\pi i/3}, 0\right), \left(e^{4\pi i/3}, 0\right), (0,1), \left(0, e^{2\pi i/3}\right), \left(0, e^{4\pi i/3}\right) \right\}$$

is a 2-design on S^3. This is visualized in Fig. 4.

Example 3.2 presents a family of 7-designs on S^3 discussed by Cohn, Conway, Elkies, and Kumar [8]. These authors also discuss the example of the D_4 root system, a 5-design on S^3 which consists of four sets of points which are each the vertices of a hexagon inscribed in a Hopf fiber. These Hopf fibers have images under the Hopf map which are the vertices of a tetrahedron on S^2, a 2-design [4, Example 2.7].

Example 3.2 As discussed by Bannai and Bannai [4, Example 2.7], the vertices

$$O := \{(1,0), (-1,0), (0,1), (0,-1), (0,e_1), (0,-e_1)\}$$

of a regular octahedron on $S^2 \subset \mathbb{R} \times \mathbb{C}$ (pictured in Fig. 2) constitute a 3-design. We also know that the vertices V_8 of a regular 8-gon are a 7-design on S^1. The $\mathbb{C}$-Hopf setting of Theorem 1.1 then gives us that

$$X := \left\{ z_y e^{2m\pi i/8} \,\middle|\, y \in O,\ m \in \{0, \ldots, 7\} \right\}$$

is a 7-design for any base points $z_y \in \pi_{\mathbb{C}}^{-1}(y)$ $(y \in O)$. We may then use (41) to pick these base points, giving rise to a concrete family of 7-designs on S^3. Figure 5 visually describes this example.

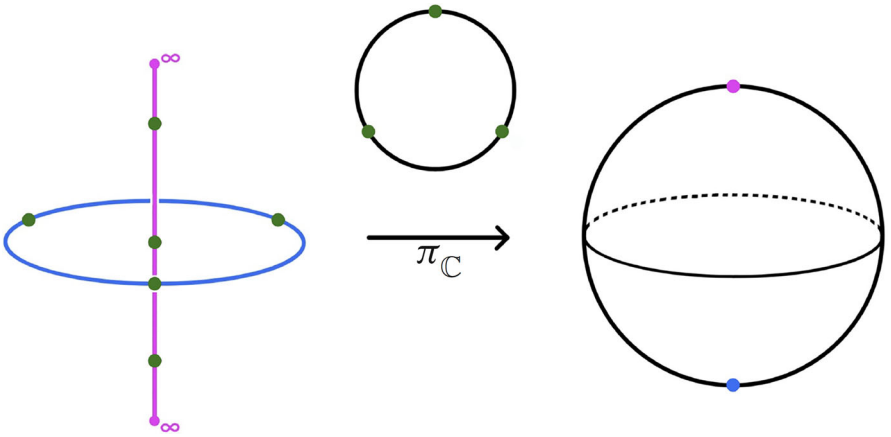


Fig. 4 A visualization of Example 3.1, where we have that $Y = \{(1, 0), (-1, 0)\} \subset S^2$ is a 1-design and each Z_y, both of which are sets of vertices of equilateral triangles, is a 2-design on S^1. Elements of X, a 2-design on $S^3 \cong \mathbb{R}^3 \cup \{\infty\}$, are shown in green on the left

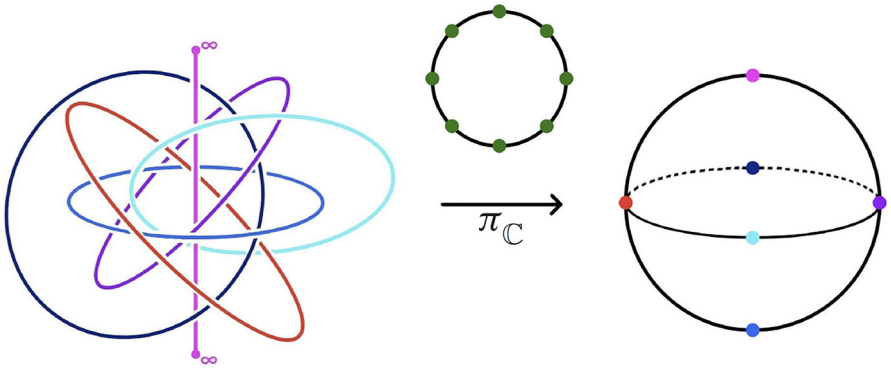


Fig. 5 The Hopf fibers in $S^3 \cong \mathbb{R}^3 \cup \{\infty\}$ corresponding to the vertices of an octahedron on S^2. To construct the family of 7-designs described in Example 3.2, distribute points corresponding to the vertices of an octagon on each marked fiber

The 24-point 5-design and 48-point 7-design discussed by Cohn, Conway, Elkies, and Kumar [8] are putatively optimally small among designs on S^3 of their respective strengths. Table 1 details when t-designs minimally small among those constructed as in the $\mathbb{C}$-Hopf setting of Theorem 1.1 have optimal size among all known t-designs on S^3.

Also note that, as Chen, Frommer, and Lang computationally verified the existence of t-designs on S^2 with $(t+1)^2$ points for $t \in \{0, ..., 100\}$ [7], we may apply the $\mathbb{C}$-Hopf setting of Theorem 1.1 to verify existence of t-designs on S^3 with $(\lfloor t/2 \rfloor + 1)^2(t+1)$ points for $t \in \{0, ..., 201\}$.

Applying Corollary 2.6, we get examples of the complex projective setting of Theorem 1.1 from Example 3.1, Example 3.2, and the cases discussed in Table 1. We now give other examples of the projective settings of the theorem.

Table 1 We here have notation as in Theorem 1.1, where Y is putatively optimally small among $\lfloor t/2 \rfloor$-designs on S^2 and we build X taking $Z_y = V_{t+1}$—an optimally small t-design on S^1 [10, Example 5.1.4]—for each $y \in Y$. The values $|Y|$ come from work of Hardin and Sloane [15, Table I] and putatively optimal minimal sizes of t-designs on S^3 were aggregated by Sloane, Hardin, and Cara [16, Table 1]. To the knowledge of the author, these designs are only known to be optimal for $t = 0$ trivially, for $\lfloor t/2 \rfloor \in \{1, 2, 3, 5\}$ on S^2 and $t \in \{1, 2, 3\}$ on S^3 by the tightness bounds of Delsarte, Goethals, and Seidel [10, Definition 5.13], and when $t = 11$ on S^3 by work of Andreev [1, Theorem 1]. While not necessarily optimal, t-designs on S^3 found by Womersley [28] may similarly indicate the efficiency of the Hopf construction of Theorem 1.1 for other values of t

t	$\lvert Y\rvert$	$\lvert X\rvert$	Putatively optimal minimum
0	1	1	1
1	1	2	2
2	2	6	5
3	2	8	8
4	4	20	20
5	4	24	24
6	6	42	42
7	6	48	48
8	12	108	96
11	12	144	120

With $\mathbb{K}$, k, n, and d as in (4) and

$$B^d := \{(1, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)\} \subset S^d,$$

$\Pi_{\mathbb{K}}(B^d)$ is a 1-design on $\mathbb{KP}^n$ for $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$. This can be seen (when $\mathbb{K} \in \{\mathbb{C}, \mathbb{H}\}$) from the fact that a set $\{[w_1], \ldots, [w_N]\}$ is a 1-design on $\mathbb{KP}^n$ if

$$\frac{1}{N} \sum_{i=1}^{N} w_i \overline{w}_i^t = \frac{1}{n+1} I_{n+1},$$

where I_{n+1} is the identity on $\mathbb{K}^{n+1}$ [9, Section 2].

Example 3.3 As $\{\pm 1\} = S^0$ is (trivially) a t-design on S^0 for any t,

$$X^n := \left\{ \pm b \;\middle|\; b \in B^n \right\} = \{(\pm 1, 0, \ldots, 0), (0, \ldots, 0, \pm 1)\}$$

is a 3-design on S^n by the real projective setting of Theorem 1.1.

For $\mathbb{K} \in \{\mathbb{C}, \mathbb{H}, \mathbb{O}\}$, X^d may similarly be built from the 1-design $\Pi_{\mathbb{K}}(B^d) \subset \mathbb{KP}^n$ and the 3-design $X^k \subset S^k$ using the $\mathbb{K}$-projective construction of Theorem 1.1. The sets X^m are known to be *tight* 3-designs [4], where a t-design is called tight if it achieves the lower bound on the size of such a design provided by Delsarte, Goethals, and Seidel [10, Definition 5.13] in the spherical setting or the analogous bounds of Dunkl [11] and of Bannai and Hoggar [5, Proposition 1.1] in the projective setting.

A number of tight projective t-designs were presented by Hoggar [17, Table 3] and Levenshtein compiled these and other projective designs into a list [21, Table 9.2]. Tight designs outlined by Hoggar include a 40-point 3-design on $\mathbb{CP}^3$ [17, Example 6], a 126-point 3-design on $\mathbb{CP}^5$ [17, Example 7], and a 165-point 3-design on $\mathbb{HP}^5$ [17, Example 9]. Noting that V_{t+1} as in (40) is a tight t-design on S^1 for any $t \in \mathbb{N}_+$ [18, Theorem A] and that the 48-point 7-design on S^3 discussed by Cohn, Conway, Elkies, and Kumar [8] as an initial example of a design built as in the complex Hopf construction of Theorem 1.1 is conjectured to be an optimally small 7-design on S^3 [16, Table 1], we get from Theorem 1.1 a 320-point 7-design on S^7, a 1,008-point 7-design on S^{11}, and a 7,920-point 7-design on S^{23}. While each of these designs is putatively optimally small among 7-designs on their respective spheres built using Theorem 1.1, these designs are not themselves tight—in fact, tightness is rare among designs built on S^d as in Theorem 1.1 for $\mathbb{K} \in \{\mathbb{C}, \mathbb{H}, \mathbb{O}\}$.

Taking $\mathbb{K}$, k, n and d as in (4) alongside $m \in \mathbb{N}_+$ and $t \in \mathbb{N}$, say $N(m,t)$ and $N(\mathbb{K},n,t)$ are the minimal numbers of points required to form a t-design on S^m and on $\mathbb{KP}^n$ respectively. Work of Bondarenko, Radchenko, and Viazovska [6, Theorem 1] combined with the tightness bounds of Delsarte, Goethals, and Seidel [10, Theorems 5.11, 5.12] shows that $N(m,t) \asymp t^m$ as $t \to \infty$ and work of Etayo, Marzo, and Ortega-Cerdà [13, Theorem 2.2] combined with the tightness bounds of Dunkl [11] shows that $N(\mathbb{K},n,t) \asymp t^{d-k}$ as $t \to \infty$. This demonstrates that each construction of Theorem 1.1 is asymptotically optimal as $t \to \infty$, in the sense that an optimally small t-design X_t capable of being built on S^d by such a construction will have size

$$|X_t| = N(\mathbb{K}, n, \lfloor t/2 \rfloor) N(k,t) \asymp t^{d-k} t^k = t^d \quad \text{as} \quad t \to \infty.$$

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